CSA1701 - ARTIFICIAL INTELLIGENCE

2.Write a Python program to solve the 8-Queen problem.

Aim: To place eight queens on a chessboard so that no two queens attack each other.

Algorithm:

1. Start with an empty chessboard.
2. Place one queen in each row.
3. Check whether the column and diagonals are safe.
4. If a position is safe, place the queen and move to the next row.
5. Backtrack when no safe position is available.
6. Print the solution.

Program:

untitled (3.14.8)

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```
N = 8

def is_safe(board, row, col):
    for i in range(col):
        if board[row][i] == 1:
            return False

    i, j = row, col
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1

    i, j = row, col
    while i < N and j >= 0:
        if board[i][j] == 1:
            return False
        i += 1
        j -= 1

    return True

def solve(board, col):
    if col >= N:
        return True

    for row in range(N):
        if is_safe(board, row, col):
            board[row][col] = 1

            if solve(board, col + 1):
                return True

            board[row][col] = 0

    return False

board = [[0 for i in range(N)] for j in range(N)]

if solve(board, 0):
    for row in board:
        print(row)
else:
    print("No solution")
```

IDLE Shell 3.14.8

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```
Python 3.14.8 (tags/v3.14.8:8e6e75d, Sep 30 2026, 18:19:33) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/praji/AppData/Local/Programs/Python/Python314/2 python.py ======
[1, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 1, 0]
[0, 0, 0, 0, 1, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 1]
[0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 1, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 1, 0, 0]
[0, 0, 1, 0, 0, 0, 0, 0]
>>>
```

Result: Thus, the 8-Queen problem was solved using backtracking.